

To Know for Quiz 2

Last 3 slides of Lecture 3 (slides 8, 9 & 10 on Hb and Mb)

Lectures 4 – 8, through protein classifications and kinase superfamily/family slide.

LG Problems 4 (most of it!) through 14.

Learning Outcomes from Lecture 3:

- Describe the distinct functional roles of hemoglobin and myoglobin.
- Define prosthetic group, holoprotein, apoprotein.
- Describe the essential role that the myoglobin protein plays in heme group binding to oxygen.

Lecture 4 Learning Outcomes:

- Define isozyme/isoenzyme/isoform, recognize examples.
- Define “primary structure” and explain why it is important and ways it can be used.
- Recognize short sequence motifs.
- Define homolog, paralog, and ortholog.
- Understand why comparison of primary structures of orthologs provides evolutionary relationships among organisms.
- Describe bioinformatics and understand what AlphaFold does.
- List the three main ways the atomic-level 3D structure of proteins can be determined.
- Recognize names of scientists who played key roles in founding protein X-ray crystallography.

Lecture 5 Learning Outcomes:

Be able to describe ...

- What is light?
- What was “the phase problem”?
- What are the six categories of non-covalent interactions? Why do we care?

Lecture 6 Learning Outcomes

The Peptide Plane

- Draw the 6 atoms in the peptide plane.
- Apply the Boltzmann Distribution equation to calculate the probability of a conformation of known energy relative to the ground state.
- Identify bonds corresponding to backbone torsion angles phi, psi and omega

The Ramachandran Plot

- Explain what information is displayed in the Ramachandran Plot
- Locate the allowed regions for Gly and for Pro vs. for all other AA
- Draw the general regions corresponding to alpha-helix and beta-sheet
- Understand the meaning of calculated vs. "real protein" Ramachandran Plot
- Explain why the Ramachandran Plots for folded and unfolded proteins look similar!

Secondary structures

- Identify and draw cis and trans isomers of the peptide bond
- Draw and recognize the backbone hydrogen bonding and key features of the alpha-helix
- Draw and recognize the backbone hydrogen bonding and key features of the beta-sheet

Lecture 7 Learning Outcomes:

Beta turns

- Describe how beta turns are defined (4 amino acid residues, $i+1$ and $i+2$ ϕ, ψ torsion angles)
- Identify beta turn positions where Pro is favored or where Gly is required

Collagen

- Describe the structure of Collagen (primary, secondary, tertiary, quaternary)
- Describe collagen synthesis (what occurs inside vs. outside the cell)
- Explain link between Vitamin C deficiency and Scurvy
- Identify what amino acid sidechains undergo post-translational modification (hydroxylation, crosslinking) and where (inside vs. outside the cell) these modifications happen.

Protein Folds

- List the main categories of folds (e.g., all alpha, all beta, etc.)
- Give reasons for the limited number of unique protein folds.
- Describe what a topology diagram shows

Lecture 8 Learning Outcomes:

- Define fold, superfamily, family
- Describe the purpose of SCOP/SCOPe, what they do and what they provide